

This DIY can monitor your door, check sound levels, check if you have any letters in the mail box, check for water leakage and check for fire and set an alarm. It can also send an email alerts.

Components needed:

- Raspberry Pi
- Arduino UNO
- Arduino Ethernet shield
- Arduino prototype shield
- Reed switch
- Wireless transceiver - RFM69HW
- Temperature sensor - DH11
- Ultrasonic sensor
- PIR sensor
- Flame sensor
- Sound sensor
- Gas / smoke sensor - MQ2
- Photoresistor
- GPS module
- Servo motor

In this DIY, the Arduino board has all the sensors i.e. temperature sensor, PIR sensor etc. connected to it. The Arduino reads the environmental data and using the Ethernet shield the sensor data is passed on to the Raspberry Pi which has OpenHAB running on it.

OpenHAB allows monitoring all these sensors from your Android or iOS smartphone via the OpenHAB app. If you have specialized lights like the Philips Hue light, OpenHAB allows controlling them via the app. To automate your own home using OpenHAB, you can find detailed instruction at <http://www.instructables.com/id/Uber-Home-Automation-w-Arduino-Pi/?ALLSTEPS>.

Give these DIYs a try and let us know how you've automated your homes and don't forget to send pictures. 